

JEE MAIN REVISION MASTER: THERMODYNAMICS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 224 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Enthalpy, Hess's Law & Bond Energies	128 PYQs	Thermochemistry
Very High	Entropy & Gibbs Free Energy Spontaneity	105 PYQs	Spontaneity

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Enthalpy, Hess's Law & Bond Energies	$\Delta H = \Delta U + \Delta n_g RT$, $\Delta_r H^\circ = \sum \Delta_f H^\circ(\text{products}) - \sum \Delta_f H^\circ(\text{reactants})$
Entropy & Gibbs Free Energy Spontaneity	$\Delta G = \Delta H - T\Delta S$, $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$, $\Delta G^\circ = -RT \ln K$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Enthalpy, Hess's Law & Bond Energies (Very High Yield — 128 PYQs)

Core Principle: State functions, standard enthalpy of formation, Hess's law of constant heat summation, and bond energy calculations.

■ **Caution / JEE Pitfall:** Standard enthalpy of formation $\Delta_f H^\circ$ of pure elements in their standard reference state is strictly ZERO (e.g. $O_2(g)$, $C(\text{graphite})$).

■ **Speed Trick:** When using bond enthalpies: $\Delta H_{\text{reaction}} = \sum \text{BE}(\text{bonds broken in reactants}) - \sum \text{BE}(\text{bonds formed in products})$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II List-I Thermodynamic Process List-II Magnitude in kJ A. Work done in reversible, isothermal expansion of 2 m..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "If 3.365 g of ethanol (l) is burnt completely in a bomb calorimeter at 298.15 K, the heat produced is 99.472 kJ. The $\Delta_f H^\circ$ is..." [Ans: 3]
- [2026 Morning Shift] Q#2 (MCQ): "Arrange the following isothermal processes in order of the magnitude of the work (p-V) involved between states 1 and 2. A. Expansion in si..." [Ans: N/A]

3.2 Entropy & Gibbs Free Energy Spontaneity (Very High Yield — 105 PYQs)

Core Principle: Second and third laws of thermodynamics, entropy changes in phase transitions, and Gibbs criteria for spontaneous change.

■ **Caution / JEE Pitfall:** For an endothermic reaction ($\Delta H > 0$) with $\Delta S > 0$, the reaction is spontaneous ONLY at HIGH temperatures where $T > (\Delta H / \Delta S)$.

■ **Speed Trick:** At equilibrium: $\Delta G = 0$ implies $T_{\text{(equilibrium)}} = (\Delta H / \Delta S)$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Match List - I with List - II. Given V_1 and V_2 are initial and final volumes respectively. List - I (Isothermal process) List - II..." **[Ans: C]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : For an ideal gas, heat capacity at constant volume is always greater than the heat capacity at constant pressure..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "20.0 dm³ of an ideal gas 'X' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until pressure of the gas is 0.2 MPa. What is the work done?" **[Ans: A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Match List - I with List - II. Given V_1 and V_2 are initial and final volumes respectively. List - I (Isothermal process) List - II (Expression) A. Reversible expansion I. $q=0$ B. Free expansion II. $q=nRT \ln \frac{V_2}{V_1}$ C. Irreversible Compression III. $w=-p_{\text{ext}}(V_1-V_2)$ D. Cyclic reversible IV. $\int \frac{dq_{\text{rev}}}{T}=0$ Choose the correct answer from the options given below :

- (A) -II, B-III, C-I, D-IV
- (B) -II, B-I, C-IV, D-III
- (C) -II, B-I, C-III, D-IV **[Correct]**
- (D) -I, B-II, C-III, D-IV

Correct Answer: C

Solution Analysis: (A) Reversible isothermal expansion For an ideal gas in a reversible and isothermal expansion: $w=-nRT \ln \frac{V_2}{V_1}$ Since the process is isothermal, $\Delta U=0$ $q=-w$ Hence, $q=nRT \ln \frac{V_2}{V_1}$ Therefore, A corresponds to II. (B) Free expansion In free expansion, gas expands against zero external pressure: $p_{\text{ext}}=0$ Thus, $w=0$ and $q=0$ (Also, for an ideal gas, $\Delta U=0$.) Therefore, B corresponds to I. (C) Irreversible isothermal compression For irreversible compression (against constant external pressure): $w=-p_{\text{ext}}(V_1-V_2)$ Therefore, C corresponds to III. (D) Cyclic reversible process For a cyclic process, $\int \frac{dq_{\text{rev}}}{T}=0$ Therefore, D corresponds to IV.

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

If 3.365 g of ethanol (l) is burnt completely in a bomb calorimeter at 298.15 K, the heat produced is 99.472 kJ. The $|\Delta H_f^\circ|$ of ethanol at 298.15 K is $\times 10^2 \text{ kJ mol}^{-1}$. (Nearest integer) Given: Standard enthalpy for combustion of graphite = -393.5 kJ mol⁻¹ Standard enthalpy of formation of water (l) = -285.8 kJ mol⁻¹ Molar mass in g mol⁻¹ of C, H, O are 12, 1 and 16 respectively

Correct Answer: 3

Solution Analysis: For complete combustion of ethanol, $\text{C}_2\text{H}_5\text{OH(l)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{CO}_2\text{(g)} + 3\text{H}_2\text{O(l)}$ First, calculate moles of ethanol burnt: Molar mass of ethanol, $\text{C}_2\text{H}_5\text{OH}$: $2(12) + 6(1) + 16 = 46 \text{ g mol}^{-1}$ So, moles of ethanol: $n = (3.365 / 46) = 0.07315 \text{ mol}$ Given heat produced for 0.07315 mol ethanol is 99.472 kJ. Hence, molar heat of combustion at constant volume: $\Delta U_{\text{(comb)}} = (-99.472 / 0.07315)$ $\Delta U_{\text{(comb)}} \approx -1359.8 \text{ kJ mol}^{-1}$ Now use the relation: $\Delta H = \Delta U + \Delta n_{\text{(g)}} RT$ Here, in the combustion reaction: Gaseous reactants = 3 mol O_2